Viraat Udar

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SUMMARY

M.S. in Computer Science (CSU Fullerton, 2026). Python developer who builds data pipelines, APIs and analysis tools, with prior experience in regulated clinical-trial data and genomics research. Two peer-reviewed publications.

TECHNICAL SKILLS

Languages: Python, R, C/C++, C#, JavaScript, Bash, HTML/CSS
Data & ML: pandas, scikit-learn, CatBoost, PyTorch, TensorFlow, statsmodels, DuckDB
Backend & Infrastructure: FastAPI, REST APIs, Docker, Kubernetes, Azure, Linux/UNIX
Tools: Git, Godot, Unity, Excel (VLOOKUP, PivotTables), RAVE EDC

PROJECTS

Hierarchical Feature Selection Framework for U.S. Census Data M.S. Capstone, 2025 to 2026
Python, FastAPI, DuckDB, CatBoost, scikit-learn, statsmodels, Godot | *udargit829.github.io/US-Map-Census-Prediction*

- Built a three-stage Python pipeline (about 9,000 lines) that finds which of roughly 528 American Community Survey variables best explain a user-supplied state or county dataset, where rows (51 or fewer) are far outnumbered by features.
- Designed a FastAPI and DuckDB data server that fetches and caches about 4,200 Census variables per geography and year, serving every later stage.
- Screened features with Pearson and Spearman correlation and Benjamini-Hochberg FDR correction, then repeatedly sampled random feature subsets, scored each with 3-fold cross-validated CatBoost, and aggregated results to feature, family and pairwise-interaction scores.
- Validated on lung cancer incidence, opioid mortality and UFO sighting data. Identified population confounding in the screening stage and traced one dataset's signal collapse to a change in the source's data access.
- Built a Godot frontend to explore the results, with a live browser demo (WebAssembly).

Cancer Cell Line Analysis Toolkit UCLA Oncology Collaboration

- Developed multithreaded Python scripts to validate and analyze large cancer cell line datasets, with association tracking and filtering (details under NDA).

Genetic Marker Analysis for Invasive Species SF Estuary Invasive Spartina Project

- Ran microsatellite analysis to separate native Spartina foliosa from invasive and hybrid variants, and estimated ancestry to assess sycamore and London plane tree hybridization.

EXPERIENCE

Lead Coding Instructor, Code Ninjas (Fullerton, CA) Aug 2023 to Present

- Teach JavaScript, C#, Unity and block coding to students ages 6 to 14; explain student progress to parents and management in plain language.

Associate Data Coordinator, Edwards Lifesciences (Irvine, CA) Jun 2022 to Dec 2022

- Worked with Data Validation and Biostatistics teams on database updates and analysis; resolved data queries in RAVE EDC.
- Performed manual data reviews and database specification QC using Excel (VLOOKUP, FILTER, PivotTables).

Lead Data Manager, GenomeAdvisors (La Mirada, CA) Jan 2021 to Jun 2022

- Supported a medical device clinical trial through to its FDA submission; helped write the Data Management Plan.
- Built the Azure infrastructure for secure, FDA-compliant data storage; maintained the Trial Master File and electronic records.
- Wrote R and Python scripts for clinical data analysis; coordinated monitoring visits and sponsor communication.

PUBLICATIONS

- Association of Mitochondrial DNA Haplogroups with Pediatric Systemic Lupus Erythematosus Disease Activity, Damage Scores, and Lupus Nephritis. Udar, V. et al. Journal of Clinical Medicine, 2025.
- Low frequency mitochondrial DNA heteroplasmy SNPs in blood, retina, and [RPE+choroid] of age-related macular degeneration subjects. Atilano, S.R. et al. (Udar, V., 4th of 6). PLOS ONE, 2021.

EDUCATION

M.S. Computer Science, California State University, Fullerton May 2026
Coursework: Neural Networks (PyTorch), Operating System Design, Web Back-End Engineering, Software Engineering.

B.S. Computer Science, California State University, Fullerton Mar 2021
Certifications: ISO 13485, ISO 14971